

a 16:10 screen. The number denotes the ratio of the horizontal size of a TV screen to its vertical size. Thus a 4:3 PAL SDTV (the most common variety) uses a fixed pixel grid of 768×576 or 720×540 . Most movies, however, are shot in 16:9 format, also known as widescreen (as seen in movie theatres); some SDTV sets and almost all HDTV sets come in a 16:9 widescreen format. A small thing to note: if you are in the market for a new PC monitor, you'd do well to pick up a widescreen variety, as most forms of entertainment now come in 16:9.

Now what happens if your TV is not 16:9? The 16:9 format movie on a 4:3 TV set will be scaled down to fit the lesser width. This leads to empty areas at the top and the bottom of the TV screen, which are filled with black bars. This is known as letterboxing. Some sets might attempt to stretch and scale the image and make it fit the 4:3 area, leading to a distorted video. Others might zoom into the video instead; this hides a portion of the video content. A 4:3 format video on a 16:9 display will fill the screen from top to bottom but not from the sides, and so you will see borders on either side of the screen.

As a general rule of thumb, all HDTV sets are likely to be widescreen, but all widescreen TV sets need not be HDTV. Thus what essentially distinguishes an HDTV from an SDTV is its screen resolution. An SDTV is 480i, an EDTV is 480p, and HDTV can be either of three resolutions: 720p, 1080i, 1080p. What do these specifications mean? The number denotes the number of horizontal lines of dots that are present in the TV. Thus a 480i has 480 lines forming its screen. An HDTV resolution is 720 and above—720p, 1080i, and 1080p—since an HDTV is widescreen and 16:9, these numbers correspond to 1280×720 and 1920×1080 pixels. The “i” stands for “Interlaced,” the “p” for “Progressive.” This is a way to denote how a picture is painted on a TV screen.

Interlaced TV sets, which most of our TV sets are, divide a video image into odd and even lines of dots. Each divided image forms a field, and when the odd field and the even field are combined, they form a complete frame. An interlaced TV draws each field every 1/60 of a second to form an illusion of running 30 frames every second. The illusion, however, isn't perfect, and



HDTV: Looks good, feels great

almost everyone can perceive flicker on our standard TV sets. A progressive scan TV set takes both the fields and combines them before displaying on screen. It thus displays a complete frame every 1/60 of a second—painting a complete picture at once, to eliminate the flicker. Progressive is almost always superior to interlaced.

Beware: some TV sets offer SDTV/EDTV resolutions along with 1080i—these are not HD sets; make sure your HDTV offers at least a 720p resolution.

Almost all HD sets available in the market today are 720p/1080i; 1080p sets are a newer breed and are much costlier than a 720p set. Generally, a 720p set can also do (or is advertised) as 1080i as well.

So which one should you buy? For all practical purposes, a 720p set is good enough. All HD content—movies, over-the-air TV content, or games—is enjoyable on a 720p set. More is, of course, better, but a 1080p set is not only significantly costlier, there is little HD content available in 1080p to best utilise the set. Subjectively, for lower screen sizes or larger sitting distances from the TV screen, it becomes very difficult to qualitatively differentiate content on a 720p set from that on a 1080p. Unless you are buying a set larger than 50 inches, you need not concern yourself with the lack of 1080p resolution.

HDMI Demystified

Our usual cables need not apply to the new HD era: S-video, SCART and composite cables are not suited for HDTV viewing. They simply do not have the required bandwidth, nor do they offer the required digital fidelity. When you move on to HDTV, you will have to deal with component, DVI and HDMI cables. The latter two are digital in nature and are the ideal course to take. A component cable will carry unprotected HD content. Most of the high-definition content will be protected, and to view this as intended, you will need HDMI ports on both the source of content (unless it's over-the-air) and on the TV set. HDMI (High-Definition Multimedia Interface) is a pure digital cable that carries audio and video and is backward-compatible with DVI-D—a DVI-D source can drive an HDMI unit through a converter, but the audio will not be available. As for the content protection: HDCP (High-Bandwidth Digital Content Protection) is a form of DRM (digital rights management) that

controls content travelling along DVI-D and HDMI. High-definition video content is downgraded to DVD quality on non-HDCP compliant TV sets, when requested by the source. DVD-Audio content is restricted to DAT quality on non-HDCP digital audio outputs (analogue audio outputs have no quality limits). Thus you will need an HDTV with an HDMI port, a video source with HDMI output, and an HDMI cable to experience real HD content.

HD-DVD and Blu-ray Disc players allow movie studios to set an Image Constraint Token (ICT) flag that will only output full-resolution HD content through HDCP. If such a player is connected to a non-HDCP-enabled HDTV, and the content is flagged, the player will output a 480p signal. The good news is that most of the movies studios seem to have informally agreed to not use this flag for the near future; additionally, none of the HD games currently have any restrictions—they can be enjoyed over both component and HDMI without niggles.